Written Report Requirements

First and foremost, following the individual original work policy clearly stated at the start of the course, all of your team's project work must be of your team's own invention. If any team members used any content at all from a particular web site or previous project, or worked on the project as part of a previous milestone, course, or existing research collaboration, you must identify all your sources and/or collaborators and provide links and citation(s) to all material that contributed to your project. This includes cases where you modified or re-used a portion of a notebook or source code provided by someone else. The instructional staff reserves the right to review all submitted work for potential plagiarism. **Claiming any work of others as your own, either by direct misattribution or failure to attribute, is grounds for academic misconduct charges.** If in doubt, please contact the instructional team.

Report Formatting

Here are the formatting rules you must follow for your report:

- You must use Times New Roman or Arial for your main font, no smaller than 10 point and no larger than 12 point font size.
- Use black text on a white background.
- Single-spaced, single column is preferred.
- The report must have a title and include the names of all team members.
- The main body of the final report will typically be at least 10 pages, but must be no more than 15 pages total. This 15 page limit includes all the main narrative text, tables, and figures. References do not count toward the page limit. Reports that clearly don't respect this page limit are likely to be penalized.
- If you want to include extra material (e.g. code, supplemental results) that's useful for more detail but not critical to your main narrative, you can put it into an appendix section, which will not count towards the page limit. However, similar to the supplemental material in a submitted conference paper it would be at the instructors' discretion on whether to include the extra appendix content in their grading.
- A reference section with formal citations is required. The expectation is that you will refer to similar work and examples and provide consistent citation practices. We are not particular regarding APA vs. MLA style, as long as the formatting is consistent. The reference section will not count agains your page count.
- The report itself must be completely self-contained in a single PDF file. If there are plots in a notebook that you want to include, you need include them as figure in the main body of the report (if they are part of your main narrative) or put them in an appendix in the report (if they aren't required to satisfy the report requirements, but you want to show more details behind a result).
- You can author your report using a Jupyter notebook as long as the exported PDF report looks exactly like a normal written report that adheres to all the requirements. The "SlideDoc" format using a presentation-like series of slides in a deck is not a format we'll accept for this Milestone.
- Please number all tables and figures!

Within the **15-page** limit, you should be able to select the key visualizations or tables you want to highlight to create a complete narrative with representative results. Extra material, per the above description, can go into an report.

Report structure

The structure of the report is semi-flexible - you can include additional information and make adjustments to the order to suit your narrative flow, but at a minimum it should have the following sections.

For the initial sections of the report below (introduction, related work, etc.), if it makes sense to separate them by supervised vs unsupervised (e.g. you used separate datasets for supervised and unsupervised learning), feel free to split that content into the corresponding supervised and unsupervised sections as needed.

Introduction (5 points)

- What problem are you trying to solve?
- What impact will solving the problem have?
- What motivated you to work on it?
- Summarize your project's supervised and unsupervised methods using for the problem, and highlight any novel contributions compared to related projects.
- What are your main findings for supervised and unsupervised learning?

Related work (5 points)

Find and reference at least three examples of existing projects or studies that are most similar to your project: include 1-2 sentence descriptions for each, and 1-2 sentences summarizing how what you're proposing is different or improved compared to this existing work. This is also the place to mention if this project is an extension of a Milestone 1 or other previous course project.

For the Data Source and Feature Engineering sections below, if you use two different datasets or different feature engineering for supervised and unsupervised learning, just describe each of those in the appropriate sections of your report and we will merge the grading into the appropriate rubric sections.

Data Source(s) (5 points)

Describe the properties of the dataset (or data API service) you used. Be specific. Your information at a minimum should include but not be limited to:

- where the datasets or API resource is located,
- what formats they returned/used,
- what were the important variables contained in them,
- how many records you used or retrieved (if using an API), and
- what time periods they covered (if there is a time element)

- Describe the steps you used to get from the raw input data to the final features used w/ supervised and unsupervised ML methods.
- Explain any initial preprocessing that was required to handle noisy or missing data.
- Be sure to include a complete list of all final features (in an appendix if necessary).

Part A. Supervised Learning (35 Total)

Methods description (8 points)

- Briefly describe your supervised learning workflow, the learning methods you used, and the feature representations you chose. You must justify why you chose your methods.
- You should have an adequate number and nature of methods: a minimum of three diverse model families must be described and explored (i.e. with very different underlying mechanisms, e.g. probabilistic, non-probabilistic, tree-based, instance-based, etc). This might be variable with larger or smaller teams and project specifics, with instructor permission.
- Include a description of how you did hyperparameter tuning or exploration with your models.
- Methods used must be clearly described, each with correct justification.

Supervised Evaluation (27 points total)
In this section you will provide a correct and comprehensive evaluation, analyzing the effectiveness of both your methods, and your feature representations.

- Overall results reporting (8 points)
  1. State and justify your choice of evaluation metrics used.
  2. Provide at least one overall summary of results that compares the best model from each family you used, in a clear, concise table.
  3. If comparing an evaluation metric between model families (e.g. comparing accuracy of support vector machines vs logistic regression), do not use just the result of a single training/test split: you must report the mean metric across multiple cross-validation folds (typically 5-fold CV), along with the standard deviation of the metric.

  Please see the **"Tips for Project Report" video** under "Week 6 Project Check-in" for key methods to get insight into your machine learning model and how to report results.
- In-Depth Evaluation (14 points)
  For the following parts of the evaluation, typically you will do these deeper analysis steps only on your best-performing model (not all of them).
  1. Feature Importance and Ablation Analysis (6 points)
     Perform a feature importance ***and*** ablation analysis on your best model to get insight into which features are or are not contributing to prediction success/failure.
  2. Sensitivity Analysis (4 points)
     Do at least one sensitivity analysis on your best model: How sensitive are your results to choice of (hyper-)parameters, features, or other varying solution elements?

4. See Advanced Model Evaluation and Failure Analysis resource for more tips

- Failure analysis (5 points)
    1. Select at least 3 ***specific*** examples **(data records)** where prediction failed, and analyze possible reasons why.
    2. Ideally you should be able to identify at least three different categories of failure.
    3. What future improvements might fix the failures? (You do not need to implement these)
    4. See Advanced Model Evaluation and Failure Analysis resource for more tips

Part B. Unsupervised Learning (25 Total)

Methods description (10 points)

- Briefly describe your unsupervised learning workflow, the learning methods you used, and the feature representations you chose. You must justify why you chose your methods.
- You should have an adequate number and nature of methods: a minimum of two unsupervised methods must be described and explored (i.e. with very different underlying mechanisms, e.g. probabilistic, non-probabilistic, tree-based, instance-based, etc). This might be variable with larger or smaller teams and project specifics, with instructor permission.
- Include a description of how you did hyperparameter tuning or exploration with your models.
- Methods used must be clearly described, each with correct justification.

Unsupervised Evaluation (15 points)
In this section you will provide a correct and comprehensive evaluation, analyzing the effectiveness of both your methods, and your choice of feature representation.

- Overall results reporting (10 points)
    1. State and justify your choice of evaluation metrics used.
    2. Provide at least one overall summary of results that compares the best model from each family you used.
    3. To summarize your findings, include at least two visualizations (chart, plot, tag cloud, map or other graphic) for each unsupervised method used that summarize your analysis.
- Do at least one sensitivity analysis on your best model: How sensitive are your results to choice of (hyper-)parameters, features, or other varying solution elements? (5 points)

Discussion (8 points)

What did you learn from doing Part A? (4 points)

- What surprised you about your results?
- What challenges did you encounter, and how did you respond to them?
- How could you extend your solution with more time/resources?

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Ethical Considerations (4 points)

- What ethical issues could arise in providing a solution to Part A, and how could you address them?
- What ethical issues could arise in providing a solution to Part B, and how could you address them?

Statement of Work (1 point)

You must include a statement that describes the contribution that each team member made to the project.

References (5 points)

You must include a reference section at the end of your report. Use standard and consistent citation style to include references to any papers, projects, web sites, or other resources you used for your project. References do not count toward the page limit.

Project Submission (1 point)

- Link to GitHub repository with all project code, pdf file of written report, and data sources.
  Note: Files edited or added to GitHub repositories after the project deadline will result in a late penalty for the final project report.
- Link to Google Doc version of written report with comments enabled.

At the discretion of the instructor up to 10 bonus points will be awarded for especially high-quality, creative or insightful projects.

Submitting the Final Report

Please submit the following components:

1. Link to GitHub repository containing:
   - Your project report as a single PDF document
   - All source code files/scripts (Python and any other code) used for your project
   - Project data in one of the following formats:
     1. The actual data files (if 25 Mb or less)

3. A sample file containing the first 100 records (if full dataset is unavailable or too large)

Note: Your GitHub repository must either be public or you must be sure to invite your project coach so they can access your files. Files edited or added to GitHub repositories after the project deadline will result in a late penalty for the final project report.

2. A Google Doc link that points to the final report with comments enabled. Coaches may choose to utilize this to provide specific, concrete, written feedback at the conclusion of the class. Please, however, allow adequate time to receive this feedback, as instructors are grading multiple dense and complicated project reports.

As part of the grading the teaching team may attempt to reproduce your results using your code and data, and you are expected to assist with this if we request it.